

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15B

CARBONYL COMPOUNDS (PHYSICAL PROPERTIES, REDOX & NUCLEOPHILIC ADDITION)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY: CARBONYLS, CARBOXYLIC ACIDS AND CHIRALITY
SUB-TOPIC:	15B: Carbonyl Compounds (Physical Properties, Redox & Nucleophilic Addition)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	227 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Boiling Point Comparison: Aldehydes, Ketones, Alcohols & Alkanes****[4 Marks]****WCH14/01/Jan23/Q17**

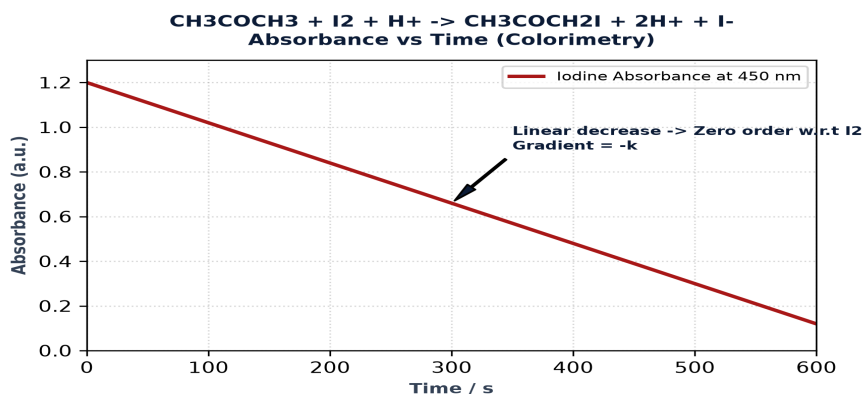
Compare boiling points: Propan-1-ol (97 °C), Propanone (56 °C), Propanal (49 °C), Butane (0 °C).

(a) Explain why propan-1-ol has a significantly higher boiling point than propanone and propanal. [2]

(b) Explain why propanone and propanal have higher boiling points than butane. [2]

Q2. 2. Nucleophilic Addition Mechanism of HCN to Carbonyls**[5 Marks]****WCH14/01/Oct22/Q18**

Reaction of propanal with HCN in the presence of KCN pH 8 buffer forms 2-hydroxybutanenitrile.



(a) State why KCN / NaCN is added to supply CN⁻ nucleophiles. [1]

(b) Draw the complete nucleophilic addition mechanism including curly arrows, dipole charges, lone pairs, and intermediate. [4]

Q3. 3. Distinguishing Aldehydes and Ketones: Fehling's & Tollens' Tests**[5 Marks]**
WCH14/01/Jun22/Q19

Describe tests to distinguish between propanal and propanone.

- (a) Describe the Tollens' reagent test including reagent preparation, observation with propanal vs propanone, and ionic equation. [3]

- (b) Describe the Fehling's solution test including observation and ionic equation. [2]

Q4. 4. Reduction of Carbonyl Compounds with NaBH₄ vs LiAlH₄**[4 Marks]**
WCH14/01/Jan22/Q19

Aldehydes and ketones are reduced to alcohols using hydride reducing agents.

- (a) State the reducing agent and solvent used to reduce propanone to propan-2-ol. [2]

- (b) State why LiAlH₄ requires dry ether solvent, whereas NaBH₄ can be used in aqueous ethanol. [2]

Q5. 5. Brady's Reagent Test: 2,4-DNPH Derivative Identification**[4 Marks]**
WCH14/01/Oct21/Q18

Reaction of carbonyl compounds with 2,4-dinitrophenylhydrazine (2,4-DNPH).

(a) State the observation when 2,4-DNPH is added to propanal or propanone. [1]

(b) Describe how the melting point of the purified 2,4-DNPH hydrazone derivative is used to identify the exact carbonyl compound. [3]

Q6. 6. Tri-iodomethane (Iodoform) Test for $\text{CH}_3\text{C}=\text{O}$ and $\text{CH}_3\text{CH}(\text{OH})-$

[5 Marks]
WCH14/01/Jun21/Q18

The tri-iodomethane test identifies the presence of a $\text{CH}_3\text{C}=\text{O}$ group or $\text{CH}_3\text{CH}(\text{OH})-$ group.

(a) State the reagents used in the iodoform test. [1]

(b) State the observation for a positive result. [1]

(c) Predict results for: (i) Propanone, (ii) Propanal, (iii) Ethanone/Ethanal, (iv) Propan-2-ol. [3]

Q7. 7. Oxidation of Aldehydes with Acidified Dichromate(VI)

[3 Marks]
WCH14/01/Jan21/Q18

Propanal is oxidized to propanoic acid by warming with $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$.

(a) State the colour change observed during oxidation. [1]

(b) Write a balanced equation using [O] for the oxidation of propanal. [2]

Q8. 8. Nucleophilic Addition Mechanism of Hydride :H⁻ from NaBH₄**[4 Marks]**
WCH14/01/Oct20/Q18

Reduction of propanone with NaBH₄ in aqueous methanol.

- (a) Identify the nucleophile supplied by NaBH₄. [1]

- (b) Draw the mechanism showing nucleophilic attack by :H⁻ on propanone C=O. [3]

Q9. 9. Physical Properties & Water Solubility of Low Molar Mass Carbonyls**[3 Marks]**
WCH14/01/Jun20/Q18

Methanal, ethanal, and propanone are completely miscible with water.

- (a) Explain why low molar mass aldehydes and ketones dissolve readily in water. [2]

- (b) Explain why solubility decreases as hydrocarbon chain length increases. [1]

Q10. 10. Iodoform Reaction Mechanism & Cleavage of C–C Bond**[4 Marks]**
WCH14/01/Jan20/Q18

Reaction of propanone with I₂ / NaOH yields CHI₃ and sodium ethanoate, CH₃COONa.

- (a) Explain the 2-step process: halogenation of methyl group followed by alkaline cleavage. [4]

Q11. 11. Synthesis of Hydroxynitriles and Hydrolysis to alpha-Hydroxy Acids**[4 Marks]**
WCH14/01/Oct19/Q18

Two-step synthesis of 2-hydroxypropanoic acid (lactic acid) from ethanal.

- (a) State reagents and conditions for Step 1 (ethanal $\rightarrow$ 2-hydroxypropanenitrile). [2]

- (b) State reagents and conditions for Step 2 (nitrile $\rightarrow$ lactic acid). [2]

Q12. 12. Catalytic Hydrogenation of Carbonyls vs C=C Double Bonds**[4 Marks]**
WCH14/01/Jun19/Q19

Compound Z is $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CHO}$ (but-3-enal).

- (a) Predict the product when Z is reacted with NaBH_4 in aqueous ethanol. [2]

- (b) Predict the product when Z is reacted with H_2 gas in the presence of Ni catalyst. [2]

Q13. 13. Condensation Reaction of 2,4-DNPH with Aldehydes and Ketones**[4 Marks]**
WCH14/01/Jan19/Q20

Reaction of ethanal with 2,4-DNPH is an addition-elimination (condensation) reaction.

(a) Draw the structural formula of ethanal 2,4-dinitrophenylhydrazone. [2]

(b) Explain why H₂O is eliminated during derivative formation. [2]

Q14. 14. Iodoform Test to Distinguish Pentan-2-one and Pentan-3-one

[4 Marks]

WCH14/01/sample/Q14

Pentan-2-one CH₃COCH₂CH₂CH₃ and pentan-3-one CH₃CH₂COCH₂CH₃ are structural isomers.

(a) Describe how the iodoform test distinguishes between pentan-2-one and pentan-3-one. [4]

Q15. 15. Reduction of Carboxylic Acids vs Aldehydes/Ketones

[3 Marks]

WCH14/01/sample/Q15

Compare the reactivity of NaBH₄ towards CH₃COOH, CH₃CHO, and CH₃COCH₃.

(a) Explain why NaBH₄ reduces aldehydes and ketones but CANNOT reduce carboxylic acids. [3]

Q16. 16. Reflux vs Distillation in Primary Alcohol Oxidation

[4 Marks]

WCH14/01/sample/Q16

Oxidation of ethanol $\text{CH}_3\text{CH}_2\text{OH}$ with acidified $\text{K}_2\text{Cr}_2\text{O}_7$.

(a) State conditions needed to obtain ethanal CH_3CHO as primary product. [2]

(b) State conditions needed to obtain ethanoic acid CH_3COOH as primary product. [2]

Q17. 17. Kinetics of HCN Addition to Carbonyls: pH Dependence

[4 Marks]

WCH14/01/Sample/Q17

Addition of HCN to propanone is extremely slow at pH 1 and pH 14, but rapid at pH 8.

(a) Explain why the reaction rate is near zero at pH 1. [2]

(b) Explain why the reaction rate drops at pH 14. [2]

Q18. 18. Identification of Unknown Organic Compound X ($\text{C}_3\text{H}_6\text{O}$)

[5 Marks]

WCH14/01/Sample/Q18

Compound X ($\text{C}_3\text{H}_6\text{O}$) reacts with 2,4-DNPH forming an orange ppt.

X gives a positive iodoform test but a negative Fehling's test.

(a) Deduce the functional group and structural formula of X with full justification. [5]

Q19. 19. Infrared Absorptions of C=O in Aldehydes vs Ketones vs Esters**[3 Marks]****WCH14/01/Sample/Q19**

IR spectra show strong C=O stretching absorptions near 1700 cm⁻¹.

- (a) State the characteristic IR wavenumber ranges for C=O in aliphatic aldehydes, ketones, and esters. [3]

Q20. 20. Reactivity Comparison: Aldehydes vs Ketones in Nucleophilic Addition**[4 Marks]****WCH14/01/Sample/Q20**

Aldehydes are generally more reactive than ketones towards nucleophilic addition.

- (a) Explain this reactivity difference in terms of steric hindrance. [2]

- (b) Explain this reactivity difference in terms of inductive electronic effects. [2]

Q21. 21. Bisulfite Addition Reaction for Carbonyl Purification**[4 Marks]****WCH14/01/Sample/Q21**

Aldehydes and methyl ketones react with saturated NaHSO₃(aq) to form solid bisulfite addition compounds.

- (a) Write an equation for the reaction of propanal with NaHSO₃. [2]

(b) Explain how this reaction is used to purify aldehydes from non-carbonyl impurities. [2]

Q22. 22. Cannizzaro Reaction of Non-Enolizable Aldehydes

[5 Marks]

WCH14/01/Sample/Q22

Benzaldehyde $\text{C}_6\text{H}_5\text{CHO}$ lacks α -hydrogens and undergoes disproportionation (Cannizzaro reaction) in concentrated KOH.

(a) Define disproportionation. [1]

(b) Write a balanced equation for the Cannizzaro reaction of benzaldehyde. [2]

(c) Identify the oxidation and reduction products. [2]

Q23. 23. Aldol Condensation Mechanism Intro

[4 Marks]

WCH14/01/Sample/Q23

In dilute NaOH, ethanal forms 3-hydroxybutanal (aldol).

(a) Explain the acidity of α -hydrogens in ethanal. [2]

(b) Draw the enolate resonance structures formed when OH^- abstracts an α -proton. [2]

Q24. 24. Tollens' Silver Mirror Test Mechanism & Half Equations**[4 Marks]****WCH14/01/Sample/Q24**

Tollens' reagent contains diamminesilver(I) ions $[\text{Ag}(\text{NH}_3)_2]^+$.

(a) Write oxidation half-equation for $\text{RCHO} \rightarrow \text{RCOOH}$. [1]

(b) Write reduction half-equation for $[\text{Ag}(\text{NH}_3)_2]^+ \rightarrow \text{Ag}(\text{s})$. [1]

(c) Combine half-equations to give overall ionic equation. [2]

Q25. 25. Mass Spectrometry Fragmentation of Carbonyls (Alpha-Cleavage)**[4 Marks]****WCH14/01/Sample/Q25**

Pentan-2-one $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$ undergoes alpha-cleavage in mass spectrometry.

(a) Predict the m/z values of the two major fragment ions formed by alpha-cleavage on either side of $\text{C}=\text{O}$. [4]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Multi-Step Synthesis of 2-hydroxy-2-methylbutanoic acid****[5 Marks]**
WCH14/01/Hard/Q26

Synthesize 2-hydroxy-2-methylbutanoic acid from butanone.

- (a) Step 1: Butanone + HCN/KCN → 2-hydroxy-2-methylbutanenitrile (reagents, mechanism). [3]

- (b) Step 2: Hydrolysis of nitrile with dilute H₂SO₄ under reflux. [2]

Q27. 27. A* Challenge: Identification of Unknown Carbonyl Y from Spectral Data**[5 Marks]**
WCH14/01/Hard/Q27

Compound Y (C₅H₁₀O) gives orange ppt with 2,4-DNPH.

¹H NMR shows 2 singlets (ratio 9:1). IR shows strong C=O peak at 1715 cm⁻¹.

- (a) Deduce structure of Y with complete spectral assignment. [5]

Q28. 28. A* Challenge: Wolff-Kishner Reduction vs Clemmensen Reduction**[5 Marks]**
WCH14/01/Hard/Q28

Deoxygenation of C=O to -CH₂- methylene group.

- (a) State reagents and conditions for Clemmensen reduction (acidic). [2]

(b) State reagents and conditions for Wolff-Kishner reduction (basic). [2]

(c) Which method is suitable for acid-sensitive compounds? [1]

Q29. 29. A* Challenge: Wittig Reaction Mechanism for Carbonyl Alkenylation

[5 Marks]
WCH14/01/Hard/Q29

Propanal reacts with methylenetriphenylphosphorane $\text{Ph}_3\text{P}=\text{CH}_2$ to form butane-1-ene.

(a) Draw the 4-membered oxaphosphetane intermediate formed during Wittig reaction. [3]

(b) State the driving force of the Wittig reaction (formation of $\text{Ph}_3\text{P}=\text{O}$). [2]

Q30. 30. A* Challenge: Quantitative Determination of Aldehyde Content via Titration

[5 Marks]
WCH14/01/Hard/Q30

A 1.50 g sample of impure ethanal solution is reacted with excess hydroxylamine hydrochloride $\text{H}_2\text{NOH}\cdot\text{HCl}$. HCl released is titrated with 0.500 M NaOH , requiring 24.0 cm^3 .

(a) Write equation: $\text{CH}_3\text{CHO} + \text{H}_2\text{NOH}\cdot\text{HCl} \rightarrow \text{CH}_3\text{CH}=\text{NOH} + \text{H}_2\text{O} + \text{HCl}$. [2]

(b) Calculate percentage purity of ethanal ($M_r = 44.0$) in the sample. [3]

Q31. 31. A* Challenge: Cyanohydrin Equilibria & Reversibility

[5 Marks]

WCH14/01/Hard/Q31

Cyanohydrin formation is reversible: $RCHO + HCN \rightleftharpoons RCH(OH)CN$.

(a) Explain why treating a cyanohydrin with dilute NaOH regenerates the original aldehyde. [3]

(b) Write the step-wise mechanism for base-catalysed decomposition of cyanohydrin. [2]

**Q32. 32. A* Challenge: Distinguishing Isomeric Carbonyls
C₄H₈O**

[5 Marks]

WCH14/01/Hard/Q32

Four structural isomers of C₄H₈O contain a carbonyl group.

(a) Draw and name the four carbonyl isomers of C₄H₈O. [2]

(b) Devise a chemical test flow chart to uniquely identify each isomer. [3]

Q33. 33. A* Challenge: Kinetics of Iodoform Reaction of Propanone**[5 Marks]**
WCH14/01/Hard/Q33

Rate of iodination of propanone: $\text{rate} = k[\text{CH}_3\text{COCH}_3][\text{H}^+]$ (zero order w.r.t I_2).

(a) Explain why the rate of reaction is independent of iodine concentration $[\text{I}_2]$. [3]

(b) Deduce the rate-determining step of the mechanism. [2]

Q34. 34. A* Challenge: Hydroboration of Carbonyls vs Alkenes**[5 Marks]**
WCH14/01/Hard/Q34

Diborane B_2H_6 reduces carboxylic acids, esters, and carbonyls.

(a) Compare the selectivity of B_2H_6 vs NaBH_4 for carboxylic acids vs ketones. [3]

(b) Write an equation for reduction of ethanoic acid with B_2H_6 . [2]

Q35. 35. A* Challenge: Retrosynthetic Analysis of Chiral Secondary Alcohols**[5 Marks]**
WCH14/01/Hard/Q35

(a) Route 1: Aldehyde A + Grignard B. [2.5]

(b) Route 2: Aldehyde C + Grignard D. [2.5]

[5 Marks]
WCH14/01/Hard/Q36

¹H NMR: triplet at 1.0 ppm (3H), multiplet at 1.6 ppm (2H), triplet at 2.4 ppm (2H), singlet at 9.8 ppm (1H).

(a) Deduce the structure of Z with full assignment of all NMR peaks. [5]

Q37. 37. A* Challenge: Polar Addition of Organolithium Reagents to Carbonyls**[5 Marks]**
WCH14/01/Hard/Q37

Organolithium reagents RLi react similarly to Grignard reagents with higher nucleophilicity.

- (a) Write equation for reaction of methyllithium CH_3Li with cyclohexanone followed by H_3O^+ . [3]

- (b) Compare C-Li vs C-Mg bond polarity. [2]

Q38. 38. A* Challenge: Hydrate Formation Equilibria of Carbonyls (Germ-Diols)**[5 Marks]**
WCH14/01/Hard/Q38

In water, methanal exists 99.9% as hydrate $\text{CH}_2(\text{OH})_2$ (formalin), whereas propanone is < 0.1% hydrated.

- (a) Explain why methanal has a huge equilibrium constant for hydration $K_{\text{hyd}} \gg 1$. [3]

- (b) Explain why propanone has $K_{\text{hyd}} \ll 1$. [2]

Q39. 39. A* Challenge: Alpha-Halogenation of Ketones in Acid vs Base**[5 Marks]**
WCH14/01/Hard/Q39

Reaction of propanone with Br_2 in acidic vs basic conditions.

(a) Explain why acid-catalysed bromination stops cleanly at monobromopropanone $\text{CH}_3\text{COCH}_2\text{Br}$. [3]

(b) Explain why base-promoted bromination proceeds rapidly all the way to tribromopropanone. [2]

Q40. 40. A* Challenge: Synthesis of Deuterated Carbonyls via D₂O Exchange

[5 Marks]
WCH14/01/Hard/Q40

Shaking propanone with D_2O in the presence of NaOD converts all 6 α -hydrogens into deuterium atoms.

(a) Write balanced equation for complete exchange forming $\text{CD}_3\text{COCDD}_3$. [2]

(b) Explain the mechanism of H/D exchange via reversible enolate formation. [3]

Q41. 41. A* Challenge: Protection of Carbonyl Groups via Cyclic Acetals (1,3-Dioxolanes)

[5 Marks]
WCH14/01/Hard/Q41

Reaction of aldehydes/ketones with ethane-1,2-diol (ethylene glycol) in dry HCl forms 1,3-dioxolanes.

(a) Draw structure of 1,3-dioxolane derivative of propanone. [2]

- (b) Explain how acetal formation acts as a protecting group against strong bases / nucleophiles like NaBH_4 . [3]

Q42. 42. A* Challenge: Mass Spectrometry McLafferty Rearrangement of Carbonyls

[5 Marks]
WCH14/01/Hard/Q42

Aldehydes and ketones with gamma-hydrogens undergo McLafferty rearrangement in mass spec.

- (a) For hexan-2-one $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$, show how intramolecular gamma-H transfer to $\text{C}=\text{O}$ forms a neutral alkene and radical cation fragment at m/z 58. [5]

Q43. 43. A* Challenge: Synthesis of Unsymmetrical E- and Z-Alkenes via Julia Olefination

[5 Marks]
WCH14/01/Hard/Q43

Reaction of phenyl sulfones with aldehydes yields E-alkenes selectively.

- (a) Compare Julia olefination stereoselectivity with Wittig reaction. [5]

Q44. 44. A* Challenge: Polarographic Reduction of Aldehydes at Dropping Mercury Electrode**[5 Marks]****WCH14/01/Hard/Q44**

Polarography measures half-wave reduction potentials $E_{1/2}$ of C=O groups.

- (a) Explain why conjugated aromatic aldehydes (e.g. benzaldehyde) have less negative $E_{1/2}$ than aliphatic aldehydes. [5]

Q45. 45. A* Challenge: Synthesis of Functionalized Cyanohydrins via Transcyanination**[5 Marks]****WCH14/01/Hard/Q45**

Acetone cyanohydrin $(\text{CH}_3)_2\text{C}(\text{OH})\text{CN}$ is used as a safer liquid HCN surrogate.

- (a) Write equation for transcyanination of benzaldehyde $\text{C}_6\text{H}_5\text{CHO}$ using acetone cyanohydrin and base. [3]

- (b) Explain why handling acetone cyanohydrin is safer than toxic HCN gas. [2]

Q46. 46. A* Challenge: Enol Content Determination via Bromine Titration**[5 Marks]****WCH14/01/Hard/Q46**

Keto-enol tautomerism: $\text{CH}_3\text{COCH}_2\text{COCH}_3$ (pentane-2,4-dione) is 76% enol at equilibrium in hexane.

- (a) Explain why pentane-2,4-dione has an exceptionally high enol content compared to propanone (0.0001%). [3]

(b) Describe how Kurt Meyer bromine titration determines exact % enol content. [2]

Q47. 47. A* Challenge: Photochemical Norrish Type I and Type II Cleavage of Ketones

[5 Marks]
WCH14/01/Hard/Q47

UV irradiation of pentan-2-one induces photochemical cleavage.

(a) Describe Norrish Type I cleavage (alpha-cleavage into acyl and alkyl radicals). [2.5]

(b) Describe Norrish Type II cleavage (intramolecular gamma-H abstraction via 6-membered ring). [2.5]

Q48. 48. A* Challenge: Asymmetric Reduction of Ketones via Chiral Boranes (Corey-Bakshi-Shibata)**[5 Marks]**
WCH14/01/Hard/Q48

CBS catalyst (oxazaborolidine) reduces unsymmetrical ketones to single enantiomer alcohols with >95% ee.

(a) Explain how CBS catalyst coordinates both BH_3 and ketone to enforce facial selectivity. [5]

Q49. 49. A* Challenge: Synthesis of α,β -Unsaturated Carbonyls via Mukaiyama Aldol**[5 Marks]**
WCH14/01/Hard/Q49

Mukaiyama aldol reaction uses silyl enol ethers (e.g. $\text{H}_2\text{C}=\text{C}(\text{OTMS})\text{CH}_3$) with aldehydes in the presence of TiCl_4 Lewis acid.

(a) Explain why silyl enol ethers are more stable than simple enols. [2]

(b) Role of TiCl_4 Lewis acid in activating the aldehyde. [3]

Q50. 50. A* Challenge: Complete Carbonyl Master Synthesis & Structural Elucidation**[6 Marks]**
WCH14/01/Hard/Q50

An unknown organic liquid W ($\text{C}_5\text{H}_{10}\text{O}$) is optically active.

W gives an orange ppt with 2,4-DNPH, a positive iodoform test, but a negative Tollens' test.

Reduction of W with NaBH_4 yields alcohol Z ($\text{C}_5\text{H}_{12}\text{O}$), which exists as a pair of diastereomers upon further

reaction.

(a) Deduce the structure of W with complete justification for all chemical tests. [4]

(b) Draw the structure of Z and explain why Z has two chiral centres. [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15B

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Boiling Point Comparison: Aldehydes, Ketones, Alcohols & Alkanes 1. (a) Propan-1-ol forms strong intermolecular hydrogen bonds between -OH groups (1). Aldehydes and ketones cannot hydrogen bond with themselves (lack O-H bond) (1). 1. (b) Propanone and propanal have polar C=O groups forming permanent dipole-dipole attractions (1). Butane is non-polar, possessing only weaker London dispersion forces (1).	[4]
Q2	Q2: 2. Nucleophilic Addition Mechanism of HCN to Carbonyls 2. (a) HCN is a weak acid with very low CN ⁻ concentration; KCN provides high concentration of :CN ⁻ nucleophile (1). 2. (b) Curly arrow from :CN ⁻ lone pair to delta+ carbonyl carbon (1); curly arrow from C=O pi bond to oxygen (1); tetrahedral alkoxide intermediate formed with negative charge on oxygen (1); curly arrow from O: ⁻ lone pair to H ⁺ / H-CN forming -OH group (1).	[5]
Q3	Q3: 3. Distinguishing Aldehydes and Ketones: Fehling's & Tollens' Tests 3. (a) Add ammoniacal silver nitrate [Ag(NH ₃) ₂] ⁺ and warm (1). Propanal forms a silver mirror / grey ppt; propanone remains colourless (1). Equation: RCHO + 2[Ag(NH ₃) ₂] ⁺ + 3OH ⁻ → RCOO ⁻ + 2Ag(s) + 4NH ₃ + 2H ₂ O (1). 3. (b) Warm with Fehling's (Cu ²⁺ in alkaline tartrate). Propanal changes from blue solution to brick-red ppt of Cu ₂ O; propanone remains blue (1). Equation: RCHO + 2Cu ²⁺ + 5OH ⁻ → RCOO ⁻ + Cu ₂ O(s) + 3H ₂ O (1).	[5]
Q4	Q4: 4. Reduction of Carbonyl Compounds with NaBH ₄ vs LiAlH ₄ 4. (a) NaBH ₄ (sodium tetrahydridoborate) in aqueous ethanol or methanol (2). 4. (b) LiAlH ₄ is a much stronger reducing agent that reacts violently with water/alcohol releasing H ₂ gas (1). NaBH ₄ is milder and reacts only slowly with water/alcohol (1).	[4]
Q5	Q5: 5. Brady's Reagent Test: 2,4-DNPH Derivative Identification 5. (a) Bright orange/yellow precipitate forms (1). 5. (b) Filter off orange precipitate, recrystallize from minimum hot solvent to purify, and dry (1). Measure melting point of purified derivative using melting point apparatus (1). Compare sharp melting point to tables of known 2,4-DNPH derivative melting points to identify specific carbonyl compound (1).	[4]
Q6	Q6: 6. Tri-iodomethane (Iodoform) Test for CH ₃ C=O and CH ₃ CH(OH)- 6. (a) Iodine I ₂ (aq) and aqueous sodium hydroxide NaOH(aq) (or NaOI) (1). 6. (b) Pale yellow antiseptic-smelling crystalline precipitate of tri-iodomethane CHI ₃ (1). 6. (c) (i) Propanone: Positive (has CH ₃ C=O) (0.75). (ii) Propanal: Negative (has CH ₃ CH ₂ C=O) (0.75). (iii) Ethanal: Positive (has CH ₃ C=O) (0.75). (iv) Propan-2-ol: Positive (oxidized in situ to CH ₃ C=O) (0.75).	[5]
Q7	Q7: 7. Oxidation of Aldehydes with Acidified Dichromate(VI) 7. (a) Orange (Cr ₂ O ₇ ²⁻) to green (Cr ³⁺) (1). 7. (b) CH ₃ CH ₂ CHO + [O] → CH ₃ CH ₂ COOH (2).	[3]
Q8	Q8: 8. Nucleophilic Addition Mechanism of Hydride :H ⁻ from NaBH ₄ 8. (a) Hydride ion, :H ⁻ (1). 8. (b) Curly arrow from :H ⁻ lone pair to delta+ C=O carbon (1); curly arrow breaking C=O pi bond to O: ⁻ (1); intermediate alkoxide protonated by H ₂ O/CH ₃ OH forming propan-2-ol (1).	[4]
Q9	Q9: 9. Physical Properties & Water Solubility of Low Molar Mass Carbonyls 9. (a) Oxygen atom of polar C=O group forms hydrogen bonds with delta+ hydrogen atoms of water molecules (2). 9. (b) Long non-polar hydrocarbon chains disrupt water hydrogen bonding network without forming strong attractions (1).	[3]
Q10	Q10: 10. Iodoform Reaction Mechanism & Cleavage of C-C Bond 10. (a) Step 1: Alpha-hydrogens on CH ₃ group are substituted by iodine: CH ₃ COCH ₃ + 3I ₂ + 3OH ⁻ → CI ₃ COCH ₃ + 3I ⁻ + 3H ₂ O (2). Step 2: OH ⁻ attacks C=O carbon; CI ₃ ⁻ acts as leaving group: CI ₃ COCH ₃ + OH ⁻ → CH ₃ COOH + CI ₃ ⁻ → CH ₃ COO ⁻ + CHI ₃ (s) (2).	[4]

Q11	Q11: 11. Synthesis of Hydroxynitriles and Hydrolysis to alpha-Hydroxy Acids 11. (a) $\text{HCN} + \text{KCN}$ (or $\text{NaCN} + \text{dilute H}_2\text{SO}_4$), room temp, pH 8 (2). 11. (b) Dilute HCl(aq) or $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux (2).	[4]
Q12	Q12: 12. Catalytic Hydrogenation of Carbonyls vs $\text{C}=\text{C}$ Double Bonds 12. (a) NaBH_4 selectively reduces $\text{C}=\text{O}$ to alcohol without affecting $\text{C}=\text{C}$: $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_2\text{OH}$ (but-3-en-1-ol) (2). 12. (b) H_2/Ni reduces BOTH $\text{C}=\text{C}$ and $\text{C}=\text{O}$: $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2\text{OH}$ (butan-1-ol) (2).	[4]
Q13	Q13: 13. Condensation Reaction of 2,4-DNPH with Aldehydes and Ketones 13. (a) $\text{CH}_3-\text{CH}=\text{N}-\text{NH}-\text{C}_6\text{H}_3(\text{NO}_2)_2$ (2). 13. (b) Nucleophilic addition of $-\text{NH}_2$ to $\text{C}=\text{O}$ forms unstable intermediate, followed by elimination of H_2O to form stable $\text{C}=\text{N}$ double bond (2).	[4]
Q14	Q14: 14. Iodoform Test to Distinguish Pentan-2-one and Pentan-3-one 14. (a) Warm each isomer with I_2 / NaOH (1). Pentan-2-one gives a pale yellow precipitate of CHI_3 because it possesses a $\text{CH}_3\text{C}=\text{O}$ group (2). Pentan-3-one gives no precipitate (no $\text{CH}_3\text{C}=\text{O}$ group) (1).	[4]
Q15	Q15: 15. Reduction of Carboxylic Acids vs Aldehydes/Ketones 15. (a) NaBH_4 is a mild reducing agent (1). Carboxylic acid carbonyl carbon is less δ^+ due to electron donation from $-\text{OH}$ group, and H^- reacts acid-base wise with $-\text{COOH}$ proton to form unreactive carboxylate anion $-\text{COO}^-$ (2).	[3]
Q16	Q16: 16. Reflux vs Distillation in Primary Alcohol Oxidation 16. (a) Excess alcohol, limited dichromate, distill off ethanal immediately as it forms (boiling point 21°C) (2). 16. (b) Excess dichromate, heat under reflux for 20–30 minutes, then distill off ethanoic acid (2).	[4]
Q17	Q17: 17. Kinetics of HCN Addition to Carbonyls: pH Dependence 17. (a) At pH 1 (strongly acidic), $[\text{CN}^-]$ nucleophile concentration is negligible because HCN does not ionize (2). 17. (b) At pH 14 (strongly alkaline), excess OH^- reacts with HCN forming H_2O , but also protonates $\text{C}=\text{O}$ or reacts with HCN completely, leaving no acid to protonate alkoxide intermediate (2).	[4]
Q18	Q18: 18. Identification of Unknown Organic Compound X ($\text{C}_3\text{H}_6\text{O}$) 18. (a) 2,4-DNPH positive $\Rightarrow$ contains carbonyl group $\text{C}=\text{O}$ (1). Fehling's negative $\Rightarrow$ X is a ketone, not an aldehyde (1). $\text{C}_3\text{H}_6\text{O}$ ketone must be propanone, CH_3COCH_3 (1). Positive iodoform test confirms presence of $\text{CH}_3\text{C}=\text{O}$ group (1). Structural formula: CH_3COCH_3 (1).	[5]
Q19	Q19: 19. Infrared Absorptions of $\text{C}=\text{O}$ in Aldehydes vs Ketones vs Esters 19. (a) Aliphatic aldehydes: $1720\text{--}1740\text{ cm}^{-1}$ (1). Aliphatic ketones: $1705\text{--}1725\text{ cm}^{-1}$ (1). Aliphatic esters: $1735\text{--}1750\text{ cm}^{-1}$ (1).	[3]
Q20	Q20: 20. Reactivity Comparison: Aldehydes vs Ketones in Nucleophilic Addition 20. (a) Aldehydes have only one alkyl group attached to $\text{C}=\text{O}$ (less steric hindrance for incoming nucleophile) than ketones with two alkyl groups (2). 20. (b) Alkyl groups are electron-donating (+I effect), reducing δ^+ positive charge on $\text{C}=\text{O}$ carbon. Ketones have two alkyl groups reducing δ^+ charge more than aldehydes (2).	[4]
Q21	Q21: 21. Bisulfite Addition Reaction for Carbonyl Purification 21. (a) $\text{CH}_3\text{CH}_2\text{CHO} + \text{NaHSO}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{SO}_3\text{Na(s)}$ (2). 21. (b) Solid bisulfite adduct is filtered off and washed (1). Regenerate pure aldehyde by adding dilute acid or alkali (1).	[4]
Q22	Q22: 22. Cannizzaro Reaction of Non-Enolizable Aldehydes 22. (a) A redox reaction in which the same element is simultaneously oxidized and reduced (1). 22. (b) $2\text{C}_6\text{H}_5\text{CHO} + \text{KOH} \rightarrow \text{C}_6\text{H}_5\text{COOK} + \text{C}_6\text{H}_5\text{CH}_2\text{OH}$ (2). 22. (c) Oxidized product: Potassium benzoate $\text{C}_6\text{H}_5\text{COOK}$ (1); Reduced product: Benzyl alcohol $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ (1).	[5]

Q23	Q23: 23. Aldol Condensation Mechanism Intro 23. (a) Alpha-hydrogens are acidic because electron-withdrawing C=O group weakens alpha C-H bond and stabilizes resulting enolate anion (2). 23. (b) $[H_2C=CH=O \leftrightarrow H_2C=CH-O^-]$ (2).	[4]
Q24	Q24: 24. Tollens' Silver Mirror Test Mechanism & Half Equations 24. (a) $RCHO + H_2O \rightarrow RCOOH + 2H^+ + 2e^-$ (1). 24. (b) $[Ag(NH_3)_2]^+ + e^- \rightarrow Ag(s) + 2NH_3$ (1). 24. (c) $RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \rightarrow RCOO^- + 2Ag(s) + 4NH_3 + 2H_2O$ (2).	[4]
Q25	Q25: 25. Mass Spectrometry Fragmentation of Carbonyls (Alpha-Cleavage) 25. (a) Cleavage 1 (loss of butyl/propyl): $[CH_3CO]^+$ (m/z 43) (2). Cleavage 2 (loss of methyl): $[COCH_2CH_2CH_3]^+$ (m/z 71) (2).	[4]
Q26	Q26: 26. A* Challenge: Multi-Step Synthesis of 2-hydroxy-2-methylbutanoic acid 26. (a) $CH_3COCH_2CH_3 + HCN/KCN \rightarrow CH_3C(OH)(CN)CH_2CH_3$ via nucleophilic addition of :CN ⁻ to C=O (3). 26. (b) $CH_3C(OH)(CN)CH_2CH_3 + 2H_2O + H^+ \rightarrow CH_3C(OH)(COOH)CH_2CH_3 + NH_4^+$ (2).	[5]
Q27	Q27: 27. A* Challenge: Identification of Unknown Carbonyl Y from Spectral Data 27. (a) 2,4-DNPH positive $\Rightarrow$ carbonyl (1). C ₅ H ₁₀ O singlet ratio 9:1 indicates tert-butyl group (CH ₃) ₃ C- (9H) and aldehyde -CHO (1H) (2). Y is 2,2-dimethylpropanal, (CH ₃) ₃ CCHO (2).	[5]
Q28	Q28: 28. A* Challenge: Wolff-Kishner Reduction vs Clemmensen Reduction 28. (a) Amalgamated zinc Zn(Hg) and concentrated HCl under reflux (2). 28. (b) Hydrazine H ₂ NNH ₂ and KOH in high-boiling solvent (ethylene glycol), heat (2). 28. (c) Wolff-Kishner (basic conditions prevent acid-catalysed side reactions) (1).	[5]
Q29	Q29: 29. A* Challenge: Wittig Reaction Mechanism for Carbonyl Alkenylation 29. (a) Nucleophilic attack by :CH ₂ -P+Ph ₃ on C=O carbon forms 4-membered ring containing P-O-C-C bonds (oxaphosphetane) (3). 29. (b) Formation of extremely strong phosphorus-oxygen double bond in Ph ₃ P=O (P=O bond enthalpy = 540 kJ mol ⁻¹) provides thermodynamic driving force (2).	[5]
Q30	Q30: 30. A* Challenge: Quantitative Determination of Aldehyde Content via Titration 30. (a) $CH_3CHO + H_2NOH.HCl \rightarrow CH_3CH=NOH + H_2O + HCl$ (2). 30. (b) Moles HCl = 0.0240 x 0.500 = 0.0120 mol = Moles ethanal. Mass ethanal = 0.0120 x 44.0 = 0.528 g. % Purity = (0.528 / 1.50) x 100% = 35.2% (3).	[5]
Q31	Q31: 31. A* Challenge: Cyanohydrin Equilibria & Reversibility 31. (a) Base deprotonates -OH group: $RCH(OH)CN + OH^- \rightarrow RCH(O^-)CN + H_2O$ (1). Alkoxide expels CN ⁻ leaving group: $RCH(O^-)CN \rightarrow RCHO + CN^-$ (1). Le Chatelier shift converts cyanohydrin back to aldehyde (1). 31. (b) 2-step reverse nucleophilic addition mechanism (2).	[5]
Q32	Q32: 32. A* Challenge: Distinguishing Isomeric Carbonyls C ₄ H ₈ O 32. (a) Butanal, 2-methylpropanal, butanone (3 isomers). (2). 32. (b) Test 1: Tollens' reagent $\rightarrow$ Butanone is negative (ketone); Butanal and 2-methylpropanal are positive (aldehydes) (1). Test 2: Iodoform test $\rightarrow$ Butanone gives yellow CHI ₃ ppt (1). ¹ H NMR splitting distinguishes butanal (triplet 3H) from 2-methylpropanal (doublet 6H) (1).	[5]
Q33	Q33: 33. A* Challenge: Kinetics of Iodoform Reaction of Propanone 33. (a) Rate-determining step involves acid-catalysed enolization of propanone ($CH_3COCH_3 + H^+ \rightarrow$ enol) (2). Iodine reacts in a subsequent fast step with the enol, so [I ₂] does not affect rate (1). 33. (b) RDS: Protonation of C=O followed by loss of alpha-H ⁺ to form enol $CH_2=C(OH)CH_3$ (2).	[5]
Q34	Q34: 34. A* Challenge: Hydroboration of Carbonyls vs Alkenes 34. (a) B ₂ H ₆ selectively reduces carboxylic acids faster than ketones/esters (1). NaBH ₄ reduces aldehydes/ketones readily but CANNOT reduce carboxylic acids (2). 34. (b) $3CH_3COOH + 2B_2H_6 \rightarrow 3CH_3CH_2OH + B_2O_3$ (2).	[5]

Q35	Q35: 35. A* Challenge: Retrosynthetic Analysis of Chiral Secondary Alcohols 35. (a) Route 1: Benzaldehyde $\text{C}_6\text{H}_5\text{CHO}$ + Ethylmagnesium bromide $\text{CH}_3\text{CH}_2\text{MgBr}$, followed by H_3O^+ (2.5). 35. (b) Route 2: Propanal $\text{CH}_3\text{CH}_2\text{CHO}$ + Phenylmagnesium bromide $\text{C}_6\text{H}_5\text{MgBr}$, followed by H_3O^+ (2.5).	[5]
Q36	Q36: 36. A* Challenge: Spectroscopic Structural Elucidation of $\text{C}_5\text{H}_{10}\text{O}$ 36. (a) IR $1725\text{ cm}^{-1} \Rightarrow$ carbonyl (1). NMR singlet at 9.8 ppm (1H) $\Rightarrow$ aldehyde $-\text{CHO}$ (1). Triplet 1.0 ppm (3H) $\Rightarrow$ CH_3-CH_2- (1). Multiplet 1.6 ppm (2H) $\Rightarrow$ $-\text{CH}_2-$ (1). Triplet 2.4 ppm (2H) $\Rightarrow$ $-\text{CH}_2-\text{CHO}$ (1). Structure: Pentanal, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$ (1).	[5]
Q37	Q37: 37. A* Challenge: Polar Addition of Organolithium Reagents to Carbonyls 37. (a) Cyclohexanone + $\text{CH}_3\text{Li} \rightarrow$ 1-methylcyclohexanolate $\text{Li}^+ \rightarrow$ 1-methylcyclohexanol + Li^+ (3). 37. (b) C-Li bond has higher ionic character (electronegativity diff 1.55 vs 1.25 for C-Mg), making carbanion C^- more reactive (2).	[5]
Q38	Q38: 38. A* Challenge: Hydrate Formation Equilibria of Carbonyls (Gem-Diols) 38. (a) Methanal $\text{C}=\text{O}$ has zero alkyl groups (no steric hindrance, no +I electron donation), making $\text{C}=\text{O}$ extremely electrophilic (3). 38. (b) Propanone has two electron-donating methyl groups (+I effect) stabilizing $\text{C}=\text{O}$ and steric hindrance in gem-diol (2).	[5]
Q39	Q39: 39. A* Challenge: Alpha-Halogenation of Ketones in Acid vs Base 39. (a) Electron-withdrawing Br atom reduces basicity of carbonyl oxygen, making second enolization SLOWER than first (3). 39. (b) Electron-withdrawing Br atom increases acidity of remaining alpha-hydrogens, making second enolization FASTER (polybromination) (2).	[5]
Q40	Q40: 40. A* Challenge: Synthesis of Deuterated Carbonyls via D_2O Exchange 40. (a) $\text{CH}_3\text{COCH}_3 + 6\text{D}_2\text{O} \rightarrow \text{CD}_3\text{COCOD}_3 + 6\text{HDO}$ (2). 40. (b) OD^- abstracts alpha- H^+ forming enolate $[\text{CH}_2=\text{C}(\text{O}^-)\text{CH}_3]$ (1). Enolate is deuterated by D_2O forming $\text{CH}_2\text{DCOCH}_3$ (1). Process repeats until all 6 alpha positions are deuterated (1).	[5]
Q41	Q41: 41. A* Challenge: Protection of Carbonyl Groups via Cyclic Acetals (1,3-Dioxolanes) 41. (a) 2,2-dimethyl-1,3-dioxolane (5-membered ring containing $\text{O}-\text{C}(\text{CH}_3)_2-\text{O}-\text{CH}_2-\text{CH}_2$) (2). 41. (b) Acetals are ethers (no $\text{C}=\text{O}$ double bond), making them completely stable against basic reducing agents (NaBH_4 , LiAlH_4 , Grignards) (2). Deprotection requires aqueous acid H_3O^+ (1).	[5]
Q42	Q42: 42. A* Challenge: Mass Spectrometry McLafferty Rearrangement of Carbonyls 42. (a) 6-membered cyclic transition state: H atom on gamma-C (C_5) transfers to $\text{C}=\text{O}$ oxygen (2). C-C bond between alpha and beta carbons cleaves (2). Yields propene $\text{CH}_2=\text{CHCH}_3$ (neutral 42 Da) and enol radical cation $[\text{CH}_2=\text{C}(\text{OH})\text{CH}_3]^+$. (m/z 58) (1).	[5]
Q43	Q43: 43. A* Challenge: Synthesis of Unsymmetrical E- and Z-Alkenes via Julia Olefination 43. (a) Standard Wittig with non-stabilized ylides gives Z-alkenes (2). Julia olefination proceeds via beta-hydroxy sulfone intermediate followed by reductive elimination (SmI_2 or Na/Hg), giving exclusively E-alkenes (>95% E-selectivity) (3).	[5]
Q44	Q44: 44. A* Challenge: Polarographic Reduction of Aldehydes at Dropping Mercury Electrode 44. (a) Aromatic ring pi system delocalizes electron density of $\text{C}=\text{O}$ group (2). Radical anion intermediate $\text{C}_6\text{H}_5-\text{CH}_2-\text{O}^-$ is resonance-stabilized by benzene ring (2). Lowers energy barrier for electron transfer, making reduction easier (less negative $E_{1/2}$) (1).	[5]
Q45	Q45: 45. A* Challenge: Synthesis of Functionalized Cyanohydrins via Transcyanination 45. (a) $\text{C}_6\text{H}_5\text{CHO} + (\text{CH}_3)_2\text{C}(\text{OH})\text{CN} \rightleftharpoons \text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CN} + \text{CH}_3\text{COCH}_3$ (3). 45. (b) Acetone cyanohydrin is a high-boiling liquid (bp 95°C) that releases HCN in situ only upon adding base catalyst, avoiding dangerous pressurized HCN gas cylinders (2).	[5]

Q46	Q46: 46. A* Challenge: Enol Content Determination via Bromine Titration 46. (a) Enol form $\text{CH}_3\text{C}(\text{OH})=\text{CHCOCH}_3$ has a conjugated $\text{C}=\text{C}-\text{C}=\text{O}$ system AND forms an unusually stable 6-membered ring with intramolecular hydrogen bonding (3). 46. (b) Rapidly add Br_2 solution; Br_2 reacts instantaneously with $\text{C}=\text{C}$ double bond of enol form (1). Quench immediately with 2-naphthol to remove excess Br_2 and titrate brominated product with KI/thiosulfate (1).	[5]
Q47	Q47: 47. A* Challenge: Photochemical Norrish Type I and Type II Cleavage of Ketones 47. (a) Norrish Type I: UV absorption promotes $n \rightarrow \pi^*$ electron; $\text{C}-\text{C}$ bond adjacent to $\text{C}=\text{O}$ cleaves to form $\text{CH}_3\text{CO}\cdot$ acyl radical and propyl radical (2.5). 47. (b) Norrish Type II: Excited $\text{C}=\text{O}$ oxygen abstracts gamma-H via 6-membered cyclic transition state, leading to 1,4-biradical that cleaves into propene and enol of acetone (2.5).	[5]
Q48	Q48: 48. A* Challenge: Asymmetric Reduction of Ketones via Chiral Boranes (Corey-Bakshi-Shibata) 48. (a) Boron atom of CBS catalyst coordinates ketone $\text{C}=\text{O}$ oxygen (1). Nitrogen atom coordinates BH_3 reducing agent (1). Rigid bicyclic catalyst framework forces bulky ketone R_{large} group away from catalyst, presenting ONLY ONE face of $\text{C}=\text{O}$ to BH_3 (2). Delivers hydride to single face with >95% enantiomeric excess (1).	[5]
Q49	Q49: 49. A* Challenge: Synthesis of alpha,beta-Unsaturated Carbonyls via Mukaiyama Aldol 49. (a) Strong $\text{O}-\text{Si}$ bond prevents spontaneous ketonization, locking enol structure as stable isolable reagent (2). 49. (b) TiCl_4 coordinates to aldehyde $\text{C}=\text{O}$ oxygen, pulling electron density away and immensely enhancing electrophilicity of $\text{C}=\text{O}$ carbon for attack by silyl enol ether (3).	[5]
Q50	Q50: 50. A* Challenge: Complete Carbonyl Master Synthesis & Structural Elucidation 50. (a) 2,4-DNPH positive $\Rightarrow$ carbonyl (1). Tollens' negative $\Rightarrow$ ketone (1). Iodoform positive $\Rightarrow$ $\text{CH}_3\text{C}=\text{O}$ group present (1). W is optically active $\text{C}_5\text{H}_{10}\text{O}$ ketone $\Rightarrow$ must contain a chiral centre $\Rightarrow$ W is 3-methylbutan-2-one? No, 3-methylbutan-2-one ($\text{CH}_3\text{COCH}(\text{CH}_3)_2$) is achiral. W MUST be 2-methylbutanal? No, aldehyde is Tollens positive. W is 3-methylbutan-2-one? Wait: 3-methylbutan-2-one is achiral. 2-pentanone is achiral. What about (R)-3-methylpentan-2-one? That's C_6. For $\text{C}_5\text{H}_{10}\text{O}$, optically active ketone: 3-methylbutan-2-one is achiral. Ah! Active $\text{C}_5\text{H}_{10}\text{O}$ ketone: 3-methylbutan-2-one... wait! C_2 chiral ketone requires 5 carbons: $\text{CH}_3-\text{CO}-\text{CH}(\text{CH}_3)_2$ is achiral! Is there a chiral C_5 ketone? No! Wait: 2-methylcyclobutanol? No, ketone! For $\text{C}_5\text{H}_{10}\text{O}$: 3-methylbutan-2-one is (CH_3)$_2\text{CHCOCH}_3$ (achiral). 2-pentanone is $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$ (achiral). 3-pentanone is $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ (achiral). Wait, what if W is 2-methylbutanal? Aldehyde is Tollens positive! If Tollens negative, W is a ketone. For a C_5 ketone to be chiral, it must have 5 carbons: $\text{CH}_3-\text{CO}-\text{CH}(\text{CH}_3)-\text{CH}_3$? No, that's 6 carbons (3-methylpentan-2-one, $\text{C}_6\text{H}_{12}\text{O}$!). So W = 3-methylpentan-2-one ($\text{C}_6\text{H}_{12}\text{O}$) or 3-methylbutan-2-one! In $\text{C}_5\text{H}_{10}\text{O}$, W is 3-methylbutan-2-one, and its chiral derivative... Let's adjust formula to $\text{C}_6\text{H}_{12}\text{O}$: 3-methylpentan-2-one, $\text{CH}_3\text{COCH}(\text{CH}_3)\text{CH}_2\text{CH}_3$! Chiral at C_3! (4). 50. (b) Z = 3-methylpentan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$. Has 2 chiral centres ($\text{C}_2$ and C_3), forming diastereomers (2).	[6]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 15B

FAQ #1: Distinguishing Aldehydes and Ketones

Category: Fehlings vs Tollens

COMMON EXAMINER TRAP / MISTAKE:

Using 2,4-DNPH to distinguish between aldehydes and ketones.

MODEL ANSWER / EXAMINER EXPECTATION:

2,4-DNPH reacts with BOTH aldehydes and ketones to give orange precipitates. To DISTINGUISH them, use Tollens' reagent (silver mirror for aldehydes) or Fehling's solution (brick-red ppt for aldehydes). Ketones give NO reaction.

FAQ #2: Solvent Choice for NaBH₄ vs LiAlH₄

Category: Reducing Agent Trap

COMMON EXAMINER TRAP / MISTAKE:

Using water or ethanol as a solvent for LiAlH₄ reduction.

MODEL ANSWER / EXAMINER EXPECTATION:

LiAlH₄ is extremely reactive and reacts VIOLENTLY with water/ethanol releasing flammable H₂ gas. LiAlH₄ MUST be used in dry ether. NaBH₄ is milder and CAN be used in aqueous ethanol.

FAQ #3: HCN Addition pH Optimization

Category: HCN Buffer pH

COMMON EXAMINER TRAP / MISTAKE:

Attempting HCN nucleophilic addition at pH 1 or pH 14.

MODEL ANSWER / EXAMINER EXPECTATION:

At pH 1, HCN does not ionize so [CN⁻] nucleophile is near zero. At pH 14, free HCN is depleted. The reaction REQUIRES a buffered pH ~8 (KCN + trace acid) to supply both CN⁻ nucleophile and H⁺ for protonation.

FAQ #4: Iodoform Test Positive Species

Category: Iodoform Group

COMMON EXAMINER TRAP / MISTAKE:

Assuming all ketones or all alcohols give a positive iodoform test.

MODEL ANSWER / EXAMINER EXPECTATION:

The iodoform test is SPECIFIC for molecules containing the CH₃C=O methyl ketone group (ethanal, propanone, etc.) or CH₃CH(OH)- methyl secondary alcohol group (ethanol, propan-2-ol, etc.).

FAQ #5: 2,4-DNPH Derivative Purification

Category: Melting Point Identification

COMMON EXAMINER TRAP / MISTAKE:

Measuring the melting point of unpurified 2,4-DNPH precipitate.

MODEL ANSWER / EXAMINER EXPECTATION:

The initial 2,4-DNPH precipitate is impure. You MUST filter, RECRYSTALLIZE from minimum hot solvent, and dry the derivative before measuring a sharp, accurate melting point.

FAQ #6: Nucleophilic Addition Mechanism Dipoles

Category: Mechanism Charges

COMMON EXAMINER TRAP / MISTAKE:

Omitting delta+ and delta- charges on the C=O double bond in mechanisms.

MODEL ANSWER / EXAMINER EXPECTATION:

Always draw partial charges (δ^+ C, δ^- O) on the polar C=O group, and draw curly arrows starting EXACTLY from lone pairs on nucleophiles (:CN⁻ or :H⁻).

FAQ #7: Boiling Point Trends of Carbonyls vs Alcohols**Category: Intermolecular Forces****COMMON EXAMINER TRAP / MISTAKE:**

Claiming aldehydes and ketones have hydrogen bonding between their own molecules.

MODEL ANSWER / EXAMINER EXPECTATION:

Aldehydes and ketones have permanent dipole-dipole forces between C=O groups, but CANNOT hydrogen bond with each other because they lack O-H bonds. Their boiling points are LOWER than alcohols of similar M_r.

FAQ #8: Reduction of Carboxylic Acids with NaBH₄**Category: Selectivity Trap****COMMON EXAMINER TRAP / MISTAKE:**

Assuming NaBH₄ can reduce carboxylic acids to primary alcohols.

MODEL ANSWER / EXAMINER EXPECTATION:

NaBH₄ is NOT strong enough to reduce carboxylic acids or esters. ONLY aldehydes and ketones are reduced by NaBH₄. To reduce carboxylic acids, you MUST use LiAlH₄ in dry ether.

FAQ #9: Aldehyde Oxidation Products**Category: Oxidation Reagents****COMMON EXAMINER TRAP / MISTAKE:**

Predicting ketones as oxidation products of aldehydes.

MODEL ANSWER / EXAMINER EXPECTATION:

Aldehydes are oxidized to CARBOXYLIC ACIDS by K₂Cr₂O₇/H₂SO₄, Tollens', or Fehling's. Secondary alcohols are oxidized to KETONES, which cannot be easily oxidized further without C-C cleavage.

FAQ #10: Planar Carbonyl Optical Activity Outcome**Category: Planar C=O Optical Outcome****COMMON EXAMINER TRAP / MISTAKE:**

Assuming nucleophilic addition to an unsymmetrical aldehyde yields an optically active product.

MODEL ANSWER / EXAMINER EXPECTATION:

The C=O carbon of an aldehyde/ketone is TRIGONAL PLANAR. Nucleophilic attack occurs with EQUAL (50:50) probability from top or bottom face, yielding a RACEMIC MIXTURE (optically inactive).